STUDENT SUMMER INTERNSHIP TECHNICAL REPORT

Development of a Robotic Vacuum Crawler for Hanford Double-Shell Tank Annuli

DOE-FIU SCIENCE & TECHNOLOGY WORKFORCE DEVELOPMENT PROGRAM

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EXECUTIVE SUMMARY

This research work has been supported by the DOE-FIU Science & Technology Workforce Development Initiative, an innovative program developed by the U.S. Department of Energy's Office of Environmental Management (DOE-EM) and Florida International University's Applied Research Center (FIU-ARC). During the summer of 2026, a DOE Fellow intern, Juan Zapata, spent 10 weeks doing a summer internship at the Hanford site and under the H2C mentor Josue Estrada company. The intern's project was initiated on Month 05, 2026, and continued through Month 08, 2026 with the objective of Developing a Robotic Vacuum Crawler for the Hanford Double-Shell Tank Annuli.

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TRADEMARKS

- Exair Line Vac 6080 is a registered trademark of EXAIR, 11510 Goldcoast Dr
- Cincinnati, OH, USA.
- Intel® RealSense™ D415 camera is a registered trademark of Intel Corporation, USA.
- Stratasys F170 3D printer is a registered trademark of Stratasys, 7665 Commerce Way
- Eden Prairie, MN USA

LIST OF ACRONYMS

- **ABS**: Acrylonitrile Butadiene Styrene
- **ASA**: Acrylonitrile Styrene Acrylate
- **CTF**: Cold Test Facility
- **CTO**: Chief Technology Office
- **DOE-EM:** Department of Energy's Office of Environmental Management
- **DSTs**: Double-Shell Tanks
- **FIU**: Florida International University
- **FIU-ARC**: Florida International University's Applied Research Center
- **H2C**: Hanford Tank Waste Operations & Closure
- **PCB**: Printed Circuit Board
- **SCFM**: Standard Cubic Feet per Minute
- **SPC**: Special Protective Coating
- **TAPI**: Tank and Pipeline Integrity
- **TEDS**: Technology Element Description Summary
- **TLR**: Technology Readiness Level
- **VCS**: Vacuum Crawler System

1.0 INTRODUCTION

1.1 Institutional Framework and Strategic Alignment

During the summer of 2026, Juan J. Zapata, a master's student in mechanical engineering from Florida International University (FIU), joined the Office of the Chief Technology Officer (CTO) at Hanford Tank Waste Operations & Closure (H2C). This research and development initiative is embedded within H2C's Tank and Pipeline Integrity (TAPI) organization, directly contributing to the U.S. Department of Energy's (DOE) overarching environmental remediation and waste management goals. This engineering effort is executed in strict alignment with the strategic objectives of the Technology and Innovation Roadmap, RPP-PLAN-43988 (Rev. 09). This roadmap formally establishes the mandate for developing automated robotic platforms and remote technologies designed for radiological surveying, inspections, and physical manipulations, with the ultimate objective of minimizing human exposure to high-radiation environments. The explicit operational criteria driving this deployment are detailed within the Technology Element Description Summary (TEDS) MTW-89. This document highlights the specific regulatory and technical requirements for the remote cleaning of concrete surfaces treated with a Special Protective Coating (SPC). Under this framework, the core focus of the current project is the systematic design, development, and refinement of a remotely operated robotic vacuum crawler system tailored specifically to navigate, vacuum, and maintain the complex internal geometries of the DST annulus space.

1.2 Overview of Hanford's Tank Infrastructure and Maintenance Challenges

The Hanford Site's 28 in-service underground Double-Shell Tanks (DSTs), constructed between 1968 and 1986, serve as critical infrastructure for the safe isolation and storage of high-level radioactive and chemical waste. Ensuring the structural integrity of these containment systems requires rigorous, routine maintenance and inspection protocols. A primary component of this effort is the periodic assessment of tank wall thinning and degradation via ultrasonic testing (UT), which must be performed on both the primary tank walls and the secondary tank bottoms.

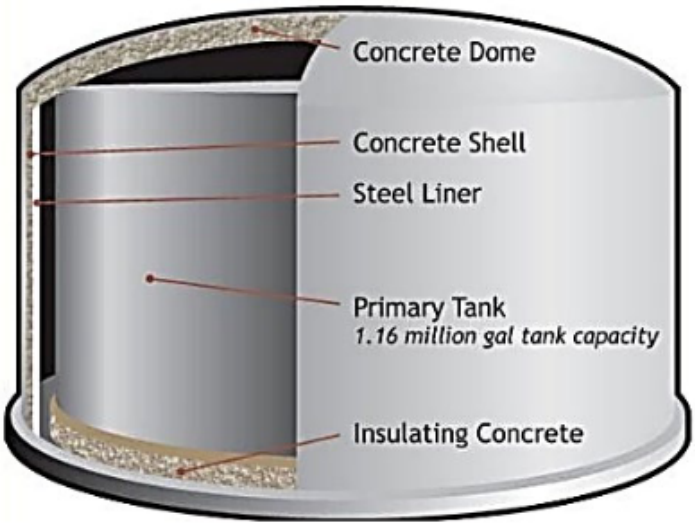

Figure 1. Underground Double-Shell Tanks constructed in 1968-1986

However, a major operational bottleneck is the accumulation of various forms of physical debris on the annulus floor. This debris—often resulting from the extensive mechanical work required to clear corrosion from the primary tank's outer surface—creates a dense barrier that interferes with and heavily hinders crucial UT inspection procedures. The urgency of addressing this debris accumulation has escalated due to recent cold spray repair operations, which have introduced a new form of waste: a fine, powdered residue that layers on top of the pre-existing debris on the annulus floor.

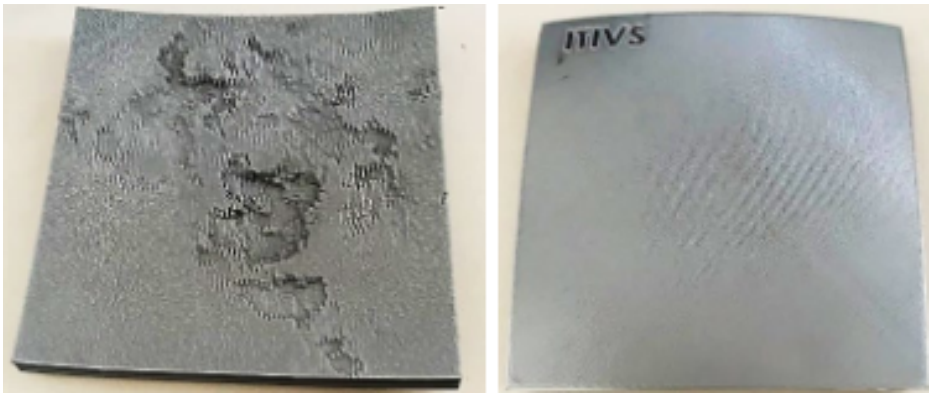

Figure 2. Representation of tank surface degradation on left and surface without degradation on right (3D printed model)

Because the annulus space is an extremely confined and radiologically hazardous environment—measuring roughly 30 inches wide with restrictive entry risers ranging from only 12 to 24 inches in diameter—manual cleanup is impossible due to severe worker exposure risks and strict physical access limits. To overcome these constraints, the Annulus Vacuum Crawler project was initiated to develop a robust, remotely operated robotic platform capable of navigating these hazardous, tightly constrained spaces to safely remove debris and clear the path for mandatory structural inspections.

Figure 3. Debris Inside Double Shell Tank AY-101

1.3 Project Overview and Objectives

Entering its third consecutive year, this project focuses on the iterative optimization, technical refinement, and software modernization of the existing Annulus Vacuum Crawler prototype. Building upon empirical performance data, field testing, and engineering feedback gathered during Year 2, the current phase transitions the platform from a proof-of-concept prototype into a robust, deployment-ready operational system. The primary objective is to enhance both the mechanical reliability and software architecture of the crawler to ensure seamless navigation and debris extraction within the highly constrained and radiologically hazardous Double-Shell Tank (DST) annulus space.

To achieve this goal, the project is structured around four core technical objectives:

- **Mechanical Refinement and Prototype Optimization:** Re-engineer and upgrade the physical crawler chassis based on Year 1 feedback to resolve operational bottlenecks. This includes optimizing the physical footprint for 12-to-24-inch riser deployment and upgrading the debris extraction interface.
- **Software Modernization and ROS2 Migration:** Overhaul the existing control software to migrate the crawler's architecture to a Robot Operating System 2 (ROS2) framework. This upgrade ensures a modular, deterministic, and scalable software stack, providing improved real-time node communication, enhanced safety critical guardrails, and standardized message packages for remote operations.
- **Optimization of Remotely Operated Control Systems:** Refine remote telemetry, live sensor feedback, and operator control interfaces. By utilizing ROS2's improved middleware, the system will deliver lower-latency data feeds to operators executing precise cleaning maneuvers from outside the high-radiation boundary.
- **Facilitation of Non-Destructive Testing (NDT):** Ensure the optimized system reliably clears a path along the primary tank walls and secondary tank bottoms. By systematically removing legacy mechanical corrosion debris and fine cold spray residue, the crawler directly enables subsequent ultrasonic testing (UT) crawlers to achieve unobstructed acoustic coupling for mandatory wall-thinning assessments.

Ultimately, this third-year phase bridges the gap between initial prototyping and field deployment—delivering a smarter, more ruggedized, ROS2-compatible robotic solution that minimizes human radiation exposure and supports Hanford's long-term environmental remediation mission.

2.0 RESEARCH DESCRIPTION

2.1 Deployment strategies

To ensure seamless integration with existing field procedures, the Year 3 operational phase maintains the baseline deployment strategy established during the initial project cycle. Engineering enhancements are designed to fit directly into the pre-existing workflow, minimizing the need for operational retraining or changes to auxiliary equipment. Crucially, all physical testing and validation are conducted at the Hanford Cold Test Facility (CTF). The CTF provides a non-radiological, full-scale mockup environment that precisely replicates the geometric and material constraints of the actual DST annulus space. This allows for rigorous, risk-free validation of the mechanical upgrades and the new ROS2 software architecture prior to actual tank deployment

Figure 4. Cold Test Facility Double- Shell Annulus Tank Deployment Mock

The deployment framework is executed across three distinct operational phases:

- **Phase I (Riser Insertion and Vertical Deployment):** The robotic crawler utilizes the exact deployment vector validated during Year 1, entering the Double-Shell Tank (DST) annulus through standard 12-to-24-inch diameter inspection risers. The system is lowered vertically to the annulus floor via a heavy-duty tether management system. This system manages the structural deployment lines, power distribution umbilical, and

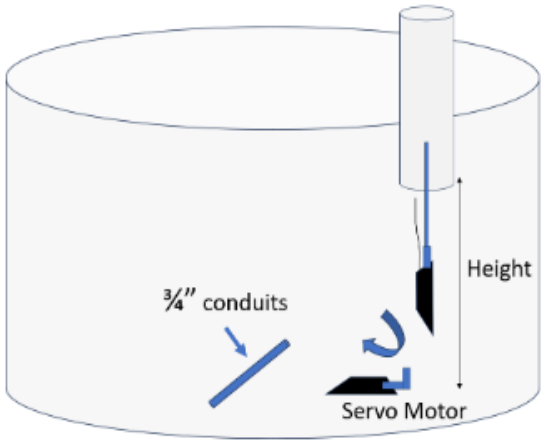

Figure 5. Diagram of the Developed Deployment Strategy

- **Phase II (Surface Transition and Annulus Floor Navigation):** Upon making contact with the 30-inch-wide annulus floor, the crawler's wheels leverage the face-down orientation to immediately establish traction. The crawler drives forward, automatically orienting itself correctly upon landing to transition to horizontal locomotion. The baseline strategy relies on navigating the curved, metallic surfaces while maintaining a precise orientation to avoid high-radiation obstacles and structural protrusions. This is where the Year 2 ROS2 software upgrade becomes critical: the introduction of a ROS2-driven control architecture allows for highly deterministic real-time communication, giving operators finer control over the crawler's traction, alignment, and velocity as it transitions from landing to operational driving.

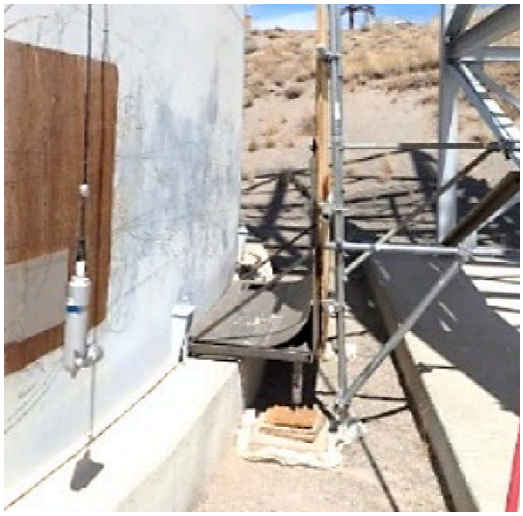

Figure 6. Deployment Mock-up

- **Phase III (Debris Extraction and Post-Inspection Clearance):** The primary functional mandate remains unchanged: the crawler systematically travels a pre-mapped cleaning path to clear legacy mechanical corrosion and fine cold spray powder residue. By adhering to the established pathing strategy but utilizing the newly optimized vacuum extraction interface and upgraded ROS2 telemetry, the platform ensures a uniformly cleared surface. This guarantees that subsequent Ultrasonic Testing (UT) crawlers can be deployed into the exact same riser and travel an unhindered path to achieve the acoustic coupling required for precise wall-thinning assessments.

Figure 7. Debris extraction early testing

2.2 Mechanical Design

Building upon empirical feedback from the Year 2 prototype, the mechanical architecture of the crawler was systematically re-engineered to resolve physical bottlenecks, simplify field maintenance, and optimize maneuverability within the tight constraints of the DST annulus. The specific hardware upgrades and structural modifications executed during this phase include:

2.21 Dimensional Reduction and Center of Gravity (CoG) Optimization

Reduced the overall length of the crawler by from 27.5in to 22.5in and the overall diameter by 1.25 inches to facilitate smoother deployment through restrictive 12-inch risers. Additionally, the Venturi vacuum system was relocated to the center of the chassis, shifting the center of gravity to the exact middle of the platform to improve stability and traction on the curved annulus floor.

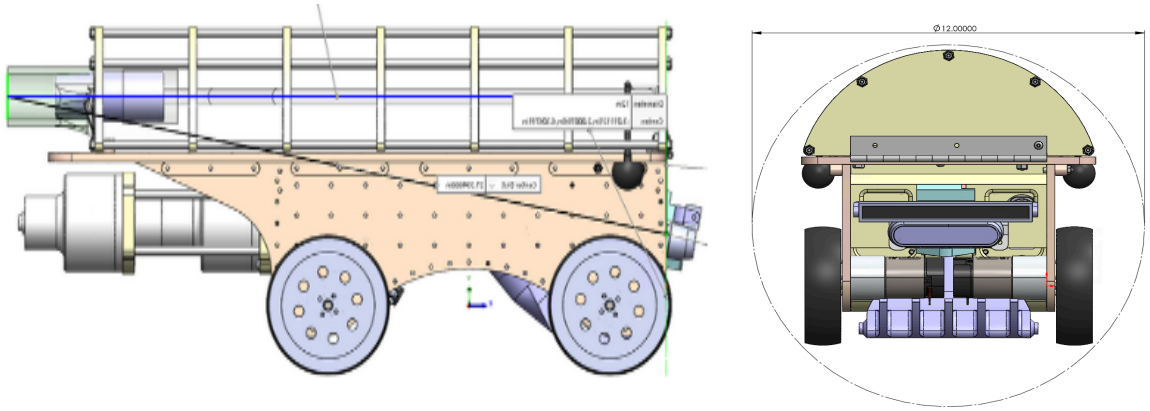

Figure 8. Vacuum Chassis Prototype V2

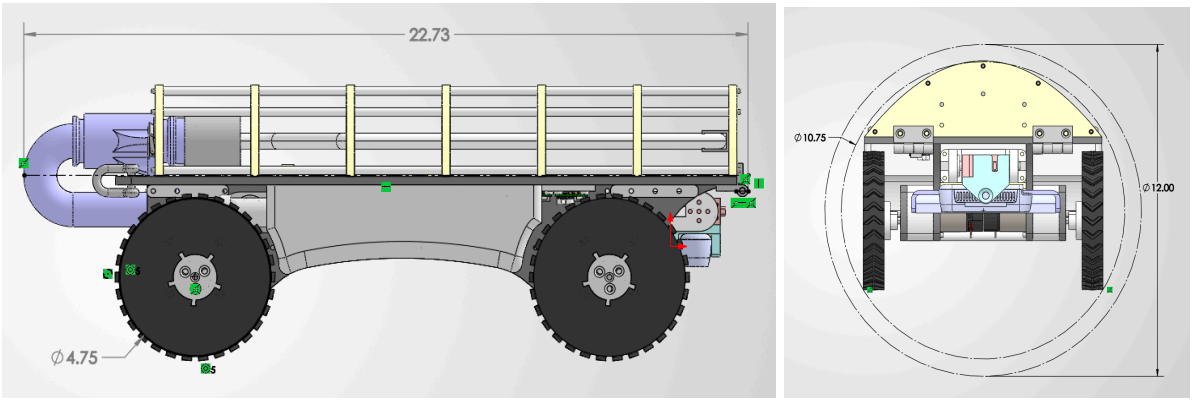

Figure 9. Side and front view for the Vacuum Chassis Prototype V3

2.22 Increased Wheel Diameter for Obstacle Clearance:

Increased the wheel diameter from 4 inches to 5 inches. This adjustment provides greater ground clearance and a better angle of approach, allowing the crawler to effortlessly roll over conduits and other structural obstacles on the annulus floor.

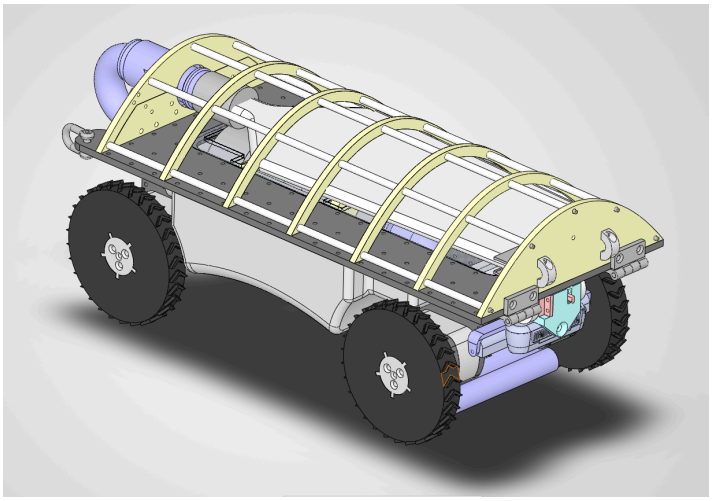

Figure 10. Side and front view for the Vacuum Chassis Prototype V3

2.23 Electronics Isolation and Maintainability : Redesign

Fully isolated the onboard electronics enclosure from the vacuum system to prevent dust contamination from the cold spray powder. The entire chassis was overhauled with a modular, maintenance-focused design to eliminate the serviceability difficulties identified in the previous iteration.

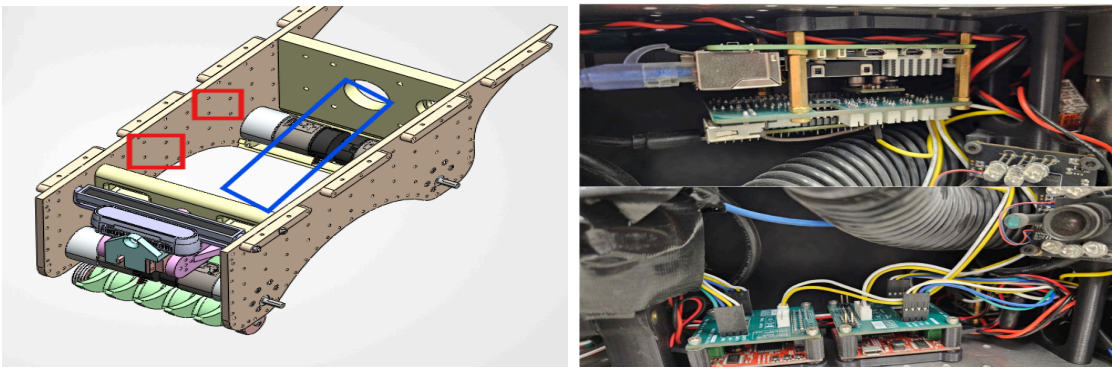

Figure 11. Vacuum Electronics distribution V2

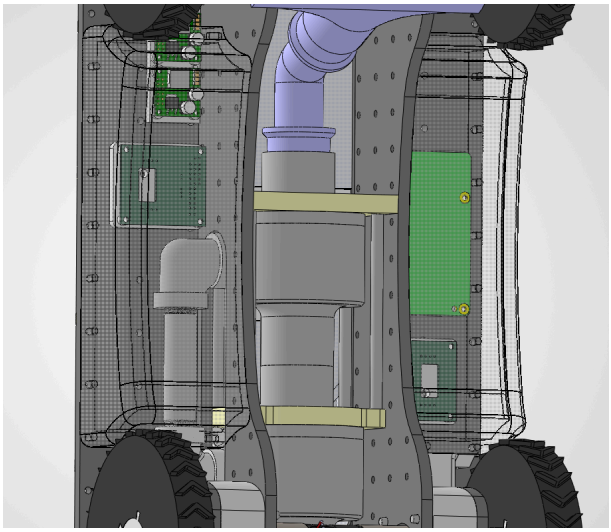

Figure 12. New Vacuum Electronics distribution V3

2.24 Modular Wheel Hub and Motor Assembly:

Developed a custom wheel hub design that decouples the drivetrain from the main chassis. This engineering change allows operators to rapidly swap out or service an electric motor without needing to disassemble the entire robotic platform, drastically reducing downtime during field operations.

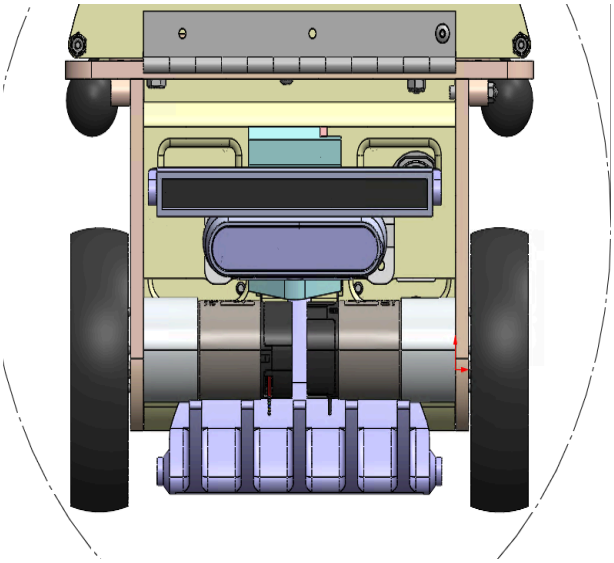

Figure 13. Vacuum wheel distribution V2

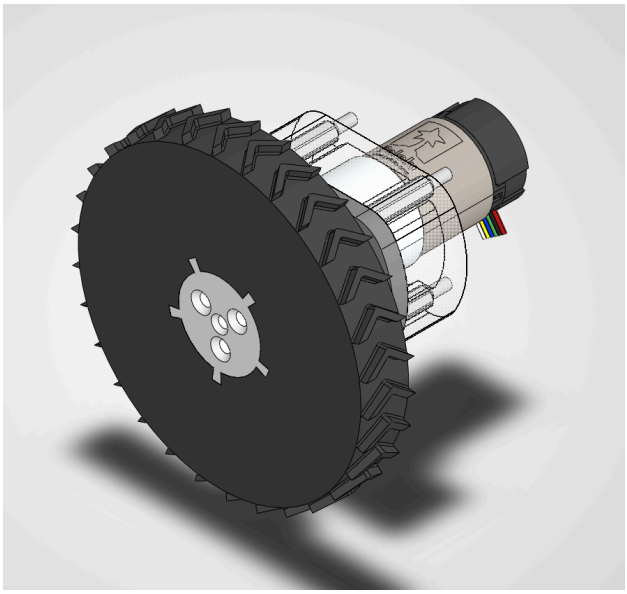

Figure 14. Vacuum wheel hub V3

2.25 Vacuum Head Geometry and Angle of Attack Optimization

To maximize aerodynamic lift and particulate disruption at the floor surface, the testing regimen evaluated different custom-designed vacuum heads. Each head geometry featured a distinct angle of attack relative to the annulus floor to determine the optimal balance between high-velocity airflow and physical debris engagement.

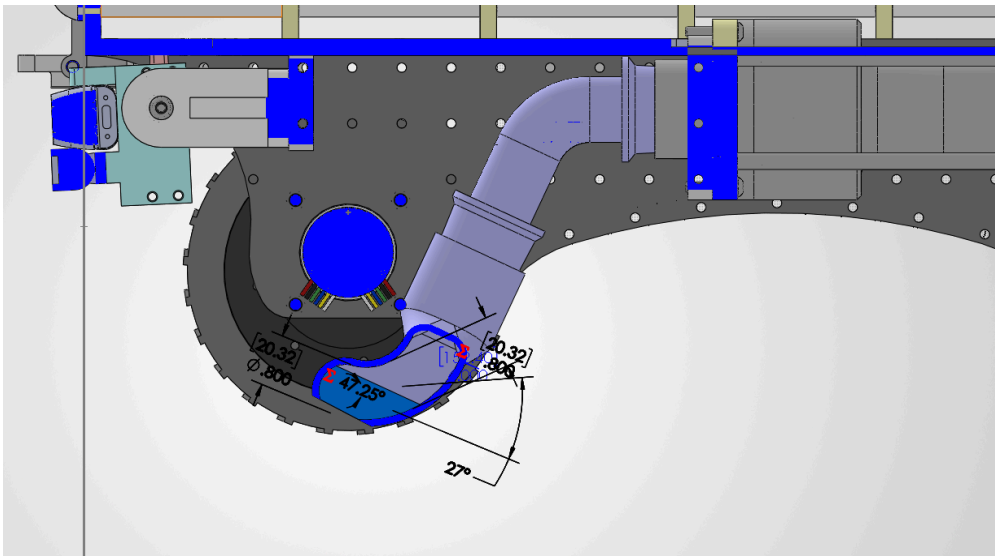

Figure 15. Vacuuming Angle of attack

2.26 TPU Cover Integration for Enhanced Traction:

Researched and implemented custom Thermoplastic Polyurethane (TPU) wheel covers. The material properties of the TPU covers maximize friction and wheel engagement against the smooth, curved steel surfaces of the tank floor, preventing slipping while navigating debris

2.3 Vacuum System Design & Development

2.31 System Architecture and Series Configuration

The baseline design, which positioned one vacuum module on the crawler and a second on the umbilical harness was superseded to eliminate pressure drops across the tether. The refined architecture integrates two EXAIR Line Vac pneumatic vacuum modules mounted directly onboard the crawler in a series configuration.

Figure 16. Sullair Air Compressor Used for Validation of the Vacuum Performance

This onboard integration was made possible by re-engineering the chassis to feature a custom CNC-machined cage, which allows the debris collection bag to be mounted directly to the platform. By positioning the modules in series and directly on the crawler, the system drastically maximizes both airflow and pull force at the cleaning nozzle, utilizing the first module to generate high primary suction and the second to accelerate the material through the exhaust.

2.32 Fluid Mechanics and Component Optimization

Pneumatic efficiency is highly sensitive to flow restrictions and pressure drops over distance. To optimize the system's fluid dynamics, several critical hardware adjustments were implemented based on empirical testing:

- Coupling and Fitting Upgrades: During development, the pneumatic coupling size was increased from 3/8 inch to 1/2 inch. This modification expanded the internal volume, minimizing the airflow restrictions and pressure losses commonly caused by standard, narrow-diameter quick-connectors.
- Hose Line Stratification: For operational demonstration, a 1-inch hose line was utilized for high-velocity material intake, which subsequently adapted to a 1/2-inch line feeding the crawler's pneumatic supply.
- Line Vac Operational Mechanics: The EXAIR modules operate by injecting compressed air through an inlet into an annular plenum chamber. The air is then throttled through directed nozzles into the throat, creating a venture-driven vacuum at the intake. Because the Line Vac features a large, smooth bore with no moving parts, it prevents clogging and provides maintenance-free conveyance of highly abrasive radiological waste over long vertical lines.

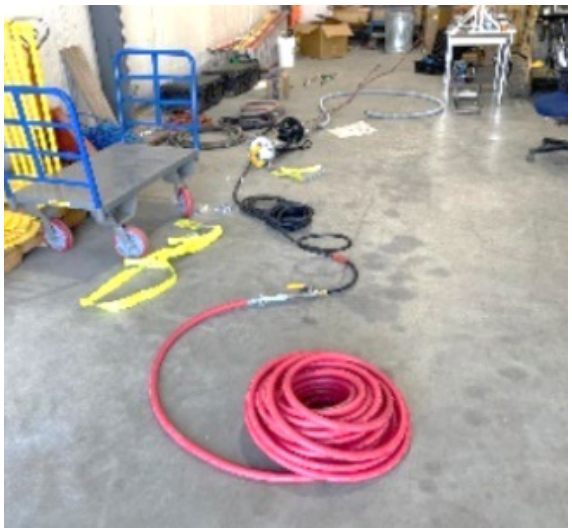

Figure 17. In Series configuration of vacuum line (1in main line to 0.5in line)

2.4 Electronics & Controls

To maintain operational continuity and hardware reliability during the Year 2 development cycle, the core electronic component stack was preserved from the previous iteration. The system continues to leverage a dual-controller architecture utilizing a Raspberry Pi as the high-level computer, a Teensy microcontroller for low-level hardware abstraction, RoboClaw motor controllers, and high-torque Pololu metal gearmotors.

Rather than altering this proven hardware foundation, the Year 2 engineering focus was strictly directed toward overhauling the system's software architecture. The previous iteration's control framework exhibited communication faults and node instability; therefore, this phase completely modernized the software ecosystem by migrating to a robust, deterministic Robot Operating System 2 (ROS2) environment while optimizing the custom physical control interface.

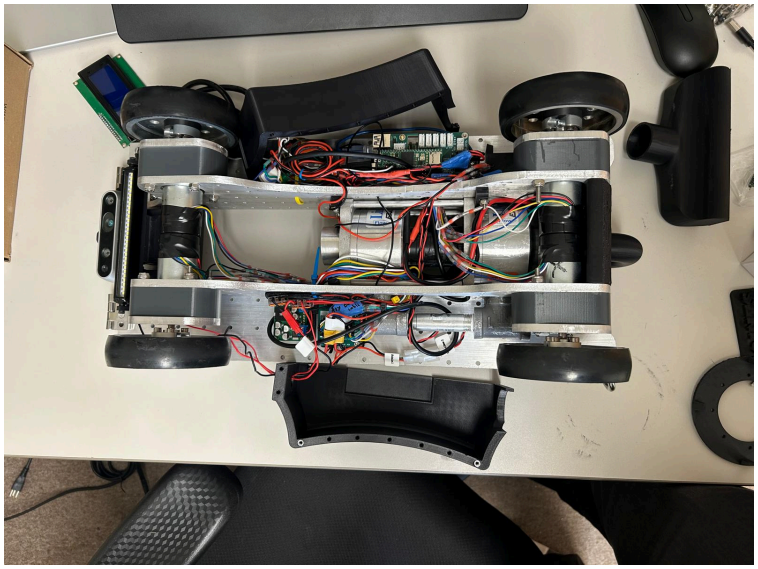

Figure 18. Electronic system of the new vacuum

2.41 Hardware Architecture & Control Interface

Signal integrity and field serviceability are maintained through a custom-designed Printed Circuit Board (PCB). This centralized PCB establishes clean, trace-routed connections between the Raspberry Pi, Teensy, and RoboClaw motor controllers. By consolidating these components onto a single board, the design eliminates erratic wiring harnesses, mitigates electromagnetic interference (EMI), and enhances overall system stability within the electromagnetically noisy environments of the tank infrastructure.

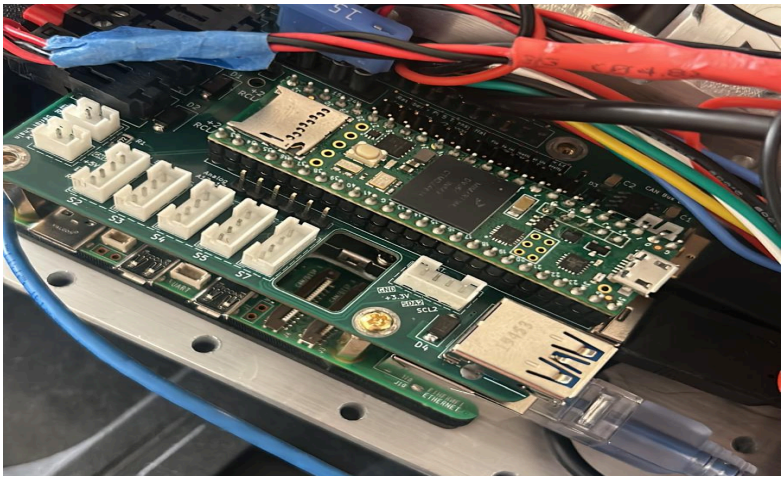

Figure 19. Raspberry pi + Teensy PCB

The operator interacts with the crawler via a professional, self-contained control box featuring an integrated touchscreen interface running an Ubuntu Linux operating system. To simplify field operations and ensure precise maneuvering, the traditional industrial joystick was replaced with an ergonomic Logitech gamepad controller. This controller allows for intuitive, simultaneous manipulation of the crawler's drivetrain, pan-and-tilt camera servo mounts, and high-intensity onboard lighting arrays. The integrated touchscreen acts as the primary human-machine interface (HMI), rendering live, real-time video feeds from the deployment zone to provide operators with critical situational awareness.

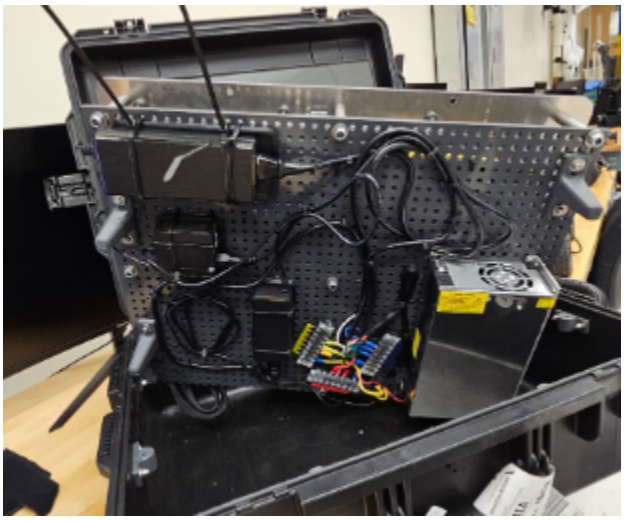

Figure 20. New Control Interface Featuring a Custom Control Box.

2.42 ROS2 Software Ecosystem & Node Architecture

The core software logic was entirely reconstructed using modular ROS2 nodes to eliminate latency bottlenecks and communication dropouts. The new control stack utilizes a decentralized topology where tasks are isolated into dedicated, predictable processes:

- Input Processing (joy and teleop_twist_joy nodes): Raw hardware inputs from the Logitech controller are captured by a standardized joy_node. These inputs are passed to a telemetry-handling node that translates button presses and analog stick deflections into precise linear and angular velocity commands (Geometry_Msgs/Twist).
- Low-Level Abstraction and Actuation: The velocity commands are ingested by the Raspberry Pi and communicated down to the Teensy microcontroller. The Teensy executes real-time motor control loops via the RoboClaw drivers to regulate the Pololu motors, while simultaneously commanding the pulse-width modulation (PWM) servos governing camera orientation.
- Dual-Camera Feed Management: A dedicated video streaming node manages the dual-camera configuration, compressing and publishing high-frame-rate visual data over the network to the operator's console without lagging the primary control loops.
- Unified Launch System: To streamline field initialization, a comprehensive ROS2 launch script was written. This single script handles node parameter configuration, establishes the Data Distribution Service (DDS) middleware layer, and activates all operational nodes concurrently, ensuring a reliable, plug-and-play startup sequence.

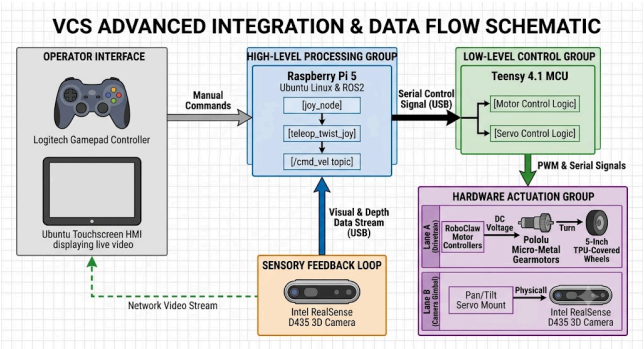

Figure 21. VCS ROS2 Planned architecture Diagram

3.0 RESULTS AND ANALYSIS

The Vacuum Crawler System (VCS) underwent a series of functional tests at the Cold Test Facility (CTF) to evaluate the performance of the mechanical, vacuum, and control-system modifications developed during this project. Testing was focused on four primary areas: deployment through the tank access riser, maneuverability within a simulated annulus environment, debris removal and vacuum performance, and validation of the ROS2 control architecture. The testing activities were designed to replicate key operational constraints associated with deployment and operation within the Double-Shell Tank (DST) annulus.

Figure 22. Control table inside CTF tank

3.1 Deployment Testing

Deployment testing was conducted to evaluate the ability of the VCS to be inserted through the restricted access geometry of a 12-inch riser and transitioned onto the annulus floor. A 12-inch riser section was used to replicate the dimensional constraint encountered during field deployment. The crawler was manually positioned and lowered through the riser by a field operator familiar with the deployment procedures used for tank operations. This test provided an operational assessment of the redesigned crawler in addition to the dimensional verification performed during the development process.

Figure 23. Deployment testing at CTF

Particular attention was given to the crawler's orientation during deployment. The previous crawler configuration incorporated a relatively flat front face, which caused the system to naturally rotate toward the wheel side as it passed through the riser and approached the annulus floor. This behavior could complicate deployment and require additional operator intervention to correctly orient the crawler. The VCS was redesigned to improve its deployment behavior and facilitate a more controlled transition from vertical insertion to horizontal operation.

3.11 Deployment Testing Result

The redesigned crawler was successfully deployed through the 12-inch riser without significant difficulty. During the test, the crawler exhibited improved orientation behavior and transitioned more predictably toward the wheel-side position upon reaching the deployment surface. Following the test, the field operator evaluated the deployment process and indicated that the redesigned VCS was easier to deploy compared with the previous year's crawler configuration. The operator feedback provided additional validation that the mechanical modifications improved the practical deployment procedure and reduced the complexity associated with positioning the crawler during insertion.

3.2 Maneuverability Testing

Maneuverability testing was performed using a full-scale simulated annulus section constructed inside the CTF. A temporary annulus structure was assembled using tarp material to reproduce the restricted operating environment between the tank walls. The simulated annulus was configured to approximately 32 inches in width, corresponding to the wall-to-wall spacing of the targeted annulus environment.

Figure 24. Deployment Mock-up at CTF

To evaluate the crawler's ability to navigate representative obstacles, obstacles were placed along the simulated annulus floor to represent the approximately 1-inch thermocouple pipes and other potential obstructions present within the operating environment. The VCS was then operated throughout the simulated annulus while performing forward and reverse motion, directional changes, and full turns. The crawler was also driven directly over and around the representative obstacles to evaluate its ability to maintain traction and continue controlled motion within the confined space.

Figure 25. Full Vacuum Deployment at CTF

3.21 Maneuverability Testing Result

The VCS successfully traversed the simulated annulus while maintaining controlled movement within the 32-inch-wide environment. The crawler completed full directional turns and successfully negotiated the simulated thermocouple obstacles without becoming immobilized. The test demonstrated that the increased wheel diameter and revised chassis configuration provided sufficient clearance and maneuverability for the representative annulus geometry. Overall, the results confirmed that the VCS was capable of performing the required driving and turning operations within the constrained environment used for testing.

3.3 Vacuum Performance and Surface Cleaning Testing

Vacuum performance and surface-cleaning testing were conducted to verify the functionality of the onboard debris-removal system. The crawler was positioned on the test surface and connected to the compressed-air supply used to operate the onboard vacuum system. Once the compressor was activated and the required pneumatic pressure was established, the vacuum system was initiated and the crawler was operated across the test surface.

Figure 26. Type of debris tested and collected

The VCS successfully generated the required suction to collect and remove debris from the test surface during operation. The crawler was driven across areas containing representative debris while the vacuum system remained active, allowing the combined mobility and debris-extraction functions to be evaluated simultaneously. The system successfully collected the material encountered during the test, demonstrating that the onboard vacuum configuration could perform surface-cleaning operations while integrated with the crawler platform.

3.31 Vacuum Performance and Surface Cleaning Testing

The successful cleaning test provided functional validation of the redesigned onboard vacuum architecture and demonstrated that the VCS could perform its intended debris-removal operation without requiring the vacuum system to remain externally mounted. These results support the use of the integrated vacuum configuration for future testing and continued development toward operation within the DST annulus environment.

3.4 ROS2 Architecture Validation

The VCS control architecture was validated using the Robot Operating System 2 (ROS2) development and visualization tools, including `rqt` and `rqt_graph`. These tools were used to verify the communication structure between the individual ROS2 nodes and confirm that the software components were properly initialized and communicating through the intended architecture.

The `rqt_graph` interface was used to visualize the active ROS2 nodes and their communication pathways during system operation. The resulting node graph confirmed the expected connections between the input, teleoperation, control, and other system processes. This verification provided evidence that the migrated ROS2 architecture was functioning as intended and that the individual software components were communicating through the defined ROS2 framework.

3.41 ROS2 Architecture Validation Result

The successful ROS2 validation confirmed the functionality of the software architecture implemented during this development phase. Combined with the physical maneuverability and deployment testing, the results demonstrate that the VCS achieved functional operation across its primary mechanical, vacuum, and software subsystems.

4.0 CONCLUSION

The third-year development of the Vacuum Crawler System (VCS) resulted in significant improvements to the crawler's mechanical configuration, deployment characteristics, debris-removal capability, and software architecture. The redesigned platform was developed to address limitations identified during previous project iterations, with particular emphasis on operation within the highly constrained geometry of the Double-Shell Tank (DST) annulus.

Functional testing at the Cold Test Facility (CTF) demonstrated that the redesigned crawler could be successfully deployed through a 12-inch riser and operated within a simulated 32-inch-wide annulus environment. Deployment testing also demonstrated improved handling and orientation during riser insertion, with field-operator feedback indicating that the new configuration was easier to deploy than the previous year's design. Maneuverability testing further demonstrated the crawler's ability to perform directional turns and negotiate representative obstacles within the confined annulus environment.

The integrated vacuum system was also successfully demonstrated during surface-cleaning tests. The crawler was able to operate the onboard vacuum system and remove representative debris from the test surface, validating the functionality of the revised vacuum configuration. In parallel, the migration to a ROS2-based control architecture was verified using ROS2 visualization and diagnostic tools, including `rqt` and `rqt_graph`. The successful communication between the system's software components confirmed the functionality of the updated control architecture.

Overall, the results demonstrate that the VCS has progressed beyond the limitations identified in previous iterations and now provides a more compact, maneuverable, and integrated platform for remote debris-removal operations. The combination of mechanical refinement, onboard vacuum integration, and ROS2-based control establishes a strong foundation for continued testing and optimization. Further development will focus on improving the crawler's reliability, debris-removal versatility, obstacle negotiation, and sensing capabilities in preparation for increasingly representative operational testing.

5.0 FUTURE WORK

Based on the feedback from operators and mentors, future development of the Vacuum Crawler System (VCS) will focus on improving its maneuverability, durability, vacuum-system versatility, obstacle avoidance, and environmental perception. The following areas have been identified as priorities for continued development:

- **Wheel Profile and Traction Optimization:** Continue evaluating different wheel profiles, tread geometries, materials, and wheel combinations to improve traction, surface engagement, stability, and maneuverability within the confined and curved DST annulus environment. Testing will focus on identifying the configuration that provides the best balance between grip, obstacle clearance, and turning performance.
- **Upward-Facing Vacuum Connection:** Redesign the vacuum hose interface so that the connection between the hose and the onboard vacuum system is oriented upward. This configuration is expected to improve hose routing, reduce interference with the crawler and surrounding structures, and provide a more suitable connection arrangement for deployment and operation within the confined annulus.
- **Manufacturing and Abrasion Protection of 3D-Printed Components:** Transition critical 3D-printed components from prototype fabrication toward more durable manufacturing methods and materials. Components exposed to continuous debris contact or abrasive operating conditions will also be evaluated for the application of protective coatings to improve wear resistance and service life.
- **Vacuum Head and Collection-Bag Optimization:** Develop and evaluate different combinations of vacuum-head geometries and debris-collection bags to accommodate varying debris types and particle sizes. This modular approach would allow the VCS to be configured according to the characteristics of the material encountered during a specific cleaning operation.
- **Vacuum Head Retention:** Improve the mechanical attachment method used to secure the 3D-printed vacuum head to the crawler. A more robust retention mechanism will be investigated to prevent the vacuum head from becoming detached during operation, particularly when the crawler encounters obstacles or uneven surfaces.

- **Adjustable Vacuum Head:** Investigate the development of a vertically adjustable vacuum head capable of moving upward and downward relative to the crawler chassis. This mechanism could improve obstacle avoidance while allowing the vacuum head to remain close to the cleaning surface during debris-removal operations.
- **LiDAR Integration:** Incorporate the LiDAR package developed and tested as part of the related sensing project into the VCS platform. The integration of LiDAR with the existing ROS2 architecture could provide enhanced environmental perception, obstacle detection, spatial awareness, and potential mapping capabilities within the DST annulus. Combining LiDAR-based sensing with the existing mobility, vacuum, and ROS2 control systems represents an important step toward a more capable and potentially semi-autonomous robotic platform.

Collectively, these developments will support the continued transition of the VCS from a functional prototype toward a more robust, adaptable, and deployment-ready robotic system capable of performing reliable debris-removal operations within the constrained DST annulus environment.

5.0 REFERENCES

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APPENDIX